

CALIBRATE BEFORE YOU RANK: CORRUPTION-VALIDATED QUALITY METRICS FOR RHYTHM-GAME CHART GENERATION

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ABSTRACT

Rhythm-game charts are designed artifacts with many valid realizations per song, yet generated charts are still evaluated with metrics borrowed from other tasks: F-measure against one reference chart, corpus language-model (LM) perplexity, and unreferenced statistics such as entropy, distinct- n , and self-similarity. We validate metrics the way one would test a smoke detector: by setting controlled fires. Applying eight targeted corruption probes at increasing doses to 167 human charts (4,175 scored variants), we find that borrowed metrics fail in ways invisible to spot checks: n -gram perplexity *improves* by 31% when charts are flattened toward the LM’s modal patterns, self-similarity structureness *rises* by 50% under copy-paste collapse, and every chart-only metric we tested (35 of 35) is completely blind to a global 60 ms audio misalignment that would be fatal in play. A two-layer protocol—raw diagnostics normalized against per-difficulty target bands calibrated on the official-chart corpus, plus an audio-anchored timing family whose anchors come from an estimated beat grid—repairs these failures: six of six targeted band scores degrade monotonically with dose (Spearman ρ -0.33 to -0.80), the band scores resist the attack that fools perplexity, and the fixed-grid timing metric recovers injected 15/30/60 ms shifts to within 0.2 ms without any reference chart. We release the probes, calibration artifacts, per-chart records, and an installable evaluation package.

1. INTRODUCTION

Automatic chart generation for rhythm games—mapping audio to a playable sequence of timed, typed notes—has progressed from onset-detection pipelines to conditional sequence models [1–4]. Evaluation has not kept pace. Because a chart is a creative artifact rather than an annotation, the same song admits many valid charts, and the field’s dominant tool—F-measure against the single official chart—penalizes every valid alternative equally and cannot be computed at all for songs without charts. The natural complement is reference-free measurement, for

which authors borrow from symbolic music and text generation: LM perplexity [1], pattern-space congruence [2], event entropy, distinct- n [5], Self-BLEU [6], feature-distribution comparison [7,8], and self-similarity structureness [9].

This paper asks whether those borrowed metrics measure chart quality at all, and answers with a validation method that requires no listening test: *targeted corruption*. Starting from human (official) charts, we apply eight damage operators—timing jitter, global anchor shift, color shuffling, copy-paste collapse, LM-argmax blandification, density scaling, burst insertion, bar shuffling—at three doses each. On average, corrupting a human chart cannot make it better, so a usable quality metric must respond in the right direction, and the response must grow with dose. Metric validation thus becomes a fully automated, reproducible procedure that requires neither human raters nor model training, and applies unchanged to any proposed chart metric.

The audit yields a negative result with a constructive core. Borrowed metrics fail structurally (Section 5.2): perplexity rewards the blandness it should punish, structureness mistakes degenerate repetition for form, several statistics are near-proxies for chart length, and all chart-only metrics are blind to global audio misalignment. But the underlying raw quantities carry signal; what is missing is calibration and grounding. We show that (i) normalizing each diagnostic against per-difficulty target bands estimated from the official corpus restores a consistent “better” direction and resists gaming, and (ii) anchoring a timing family on a beat grid *estimated from audio* closes the misalignment blind spot with no reference chart and no measurable accuracy cost.

Our contributions are:

1. **A corruption-probe audit harness** for chart-quality metrics: eight dose-controlled damage operators with acceptance criteria (monotone dose response; anti-gaming; published coupling structure) and fairness rules, applicable to any future metric (Section 4).
2. **A structural failure analysis** of 14 borrowed metrics, including two reductio exhibits—perplexity improves under blandification; structureness improves under collapse—and an all-blind result for chart-only metrics under anchor shift (Section 5).



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3. **A two-layer, reference-free evaluation protocol:** raw diagnostics plus corpus-calibrated band scores with an empirically assigned role taxonomy, extended with an audio-anchored timing family whose fixed-grid offset detector recovers injected shifts to sub-ms accuracy (Section 3, Section 5.3).
4. **Released artifacts:** an installable evaluation package, probe implementations, calibration bands, all per-chart records, and the grouped song-level dataset splits. [PENDING: P8: public repo URL.]

The probes establish what metrics detect: *damage*. Whether the surviving metrics also track player experience is a distinct question that requires perception studies; we address it in Section 6.

2. RELATED WORK

Chart generation and its evaluation. Dance Dance Convolution evaluates step placement by F-measure against a reference chart and step selection by LM perplexity [1]; later systems inherit this template [3, 4]. TaikoNation adds pattern-congruence measures against human charts [2]. None of these metrics has been audited against controlled damage.

Generative-music evaluation. Yang and Lerch compare feature distributions between generated and reference sets [7]; MusPy popularized symbolic statistics [8]; the Jazz Transformer introduced self-similarity structure indicators [9]. These are set-level or direction-ambiguous quantities; we audit their behavior when used as per-chart quality signals.

Metric critiques. Theis et al. showed that likelihood and sample quality can diverge arbitrarily [10]; our blandification result is a sharp corpus-LM instance. NLG research documented pathologies of distinct- n and Self-BLEU [5, 6]. Unlike pure critiques we supply the repair, and unlike human-anchored metric studies our harness needs no raters. Onset F-measure follows `mir_eval` [11].

3. TWO-LAYER EVALUATION PROTOCOL

Data. We use a corpus of 1,155 taiko-style chart files covering 1,013 unique songs (grouped by audio identity and normalized title), split at the song level into 813/80/120 training/validation/test songs so that no song appears in more than one split. Calibration uses official chart-courses from the training split; evaluation uses 40 held-out test songs across all difficulty courses. [PENDING: P9: refresh counts and all downstream numbers from the clean-split re-run.]

Layer 1: raw diagnostics. Each chart is a sequence of timed, typed events. We compute quantities with native units and no claimed direction: note density and local-density percentiles, inter-onset entropy, color-switch rate, n -gram LM negative log-likelihood (trigram, trained on the calibration corpus), unseen-transition rate, unique/repeated 4-gram rates, and windowed LM surprisal

Probe (dose knob)	Damages	Designated target
C1 timing jitter ($\pm 10/20/30$ ms)	micro-timing	rhythm complexity
C2 anchor shift (+15/30/60 ms)	audio alignment	timing family (Section 5.3)
C3 type shuffle ($p=.2/.4/.8$)	note grammar	transition validity
C4 loop collapse ($p=.3/.6/1$)	structure	repetition adequacy
C5 blandification ($p=.3/.6/1$)	interestingness	(anti-gaming exhibit)
C6 density scale ($\times .5/1.5/2$)	difficulty fit	density adequacy
C7 burst insert (2/4/8 clusters)	playability	density-spike score
C8 bar shuffle ($p=.3/.6/1$)	global form	repetition adequacy

Table 1. Eight corruption probes applied to 167 official charts at three doses each (4,175 scored variants incl. originals; deterministic seeds; operators verified by independent re-execution). Every probe preserves what it does not target: C3–C5 preserve all note times exactly; C4 and C6 preserve or control density.

percentiles. These deliberately coincide with the borrowed metrics: the audit compares *normalization strategies*, not measurement choices.

Layer 2: calibrated band scores. For each diagnostic and difficulty course we estimate a target band $[p_{10}, p_{90}]$ over the calibration corpus. Scores are 1 inside the band and decay smoothly outside; error-type quantities use exponential penalties. This resolves direction ambiguity by construction—“more entropy” is neither good nor bad, but “entropy within the range human charters use at this difficulty” is well defined—and it is reference-free at the song level: scoring a chart for a novel song requires only the fixed corpus calibration.

Role taxonomy. Scores are assigned roles empirically by the audit, not a priori: *rankers* respond to damage and separate systems (pattern-IC, transition validity, repetition and variety adequacy, rhythm complexity); *constraints* saturate near 1 for all reasonable charts and flag only gross violations (density, strain, overload); *diagnostics* carry signal but immature calibration (the current boredom heuristic; the hypersensitive rare- n -gram rate). We report score profiles, not a single aggregate: fixing trade-off weights between axes is exactly the judgment we defer to human data.

4. THE CORRUPTION-PROBE HARNESS

Table 1 lists the probes. Corruption direction is guaranteed at the population level: since band scores are two-sided, a corruption can push an individual off-band chart *toward* its band (measured: 4–14% of charts per probe at max dose), so all criteria are evaluated on means over 167 charts. Acceptance requires (a) *sensitivity*: the designated target score decreases monotonically with dose; (b) *anti-gaming*: no score family should reward damage; and (c) *disclosed coupling*: the full probe $\times$ metric response matrix is published rather than summarized by its diagonal.

Fairness rules fixed in advance: identical probes for baselines and calibrated scores; identity pairs across layers (distinct-4 $\equiv$ unique-4-gram rate; transition validity $\leftrightarrow$ unseen-transition rate) are never counted as independent agreement; length confounds are checked against note

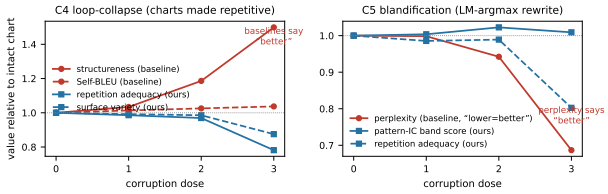


Figure 1. Borrowed metrics reward damage; calibrated scores respond correctly. Values are relative to the intact chart (dotted line = no change); red = borrowed baselines, blue = calibrated scores. **Left (C4 loop collapse):** replacing charts with their own most common 4-gram loop raises structureness by 50% ($0.096 \rightarrow 0.144$) and Self-BLEU from 0.860 to 0.892—both read as improvement—while repetition and variety adequacy correctly fall. **Right (C5 blandification):** rewriting notes to the LM argmax lowers perplexity by 31% ($7.98 \rightarrow 5.48$), i.e. “better” under the standard reading, while the pattern-IC band score refuses to improve and repetition adequacy catches the induced looping. The same reversal holds when the attack is driven by an LM trained on a disjoint corpus half ($7.98 \rightarrow 5.55$).

count; and the LM’s provenance is audited (below), with the blandification attack additionally run *cross-LM* (drive and measure LMs trained on disjoint corpus halves).

5. RESULTS

5.1 Sensitivity: targeted damage moves the right scores

All six probes with calibrated targets pass: dose–response Spearman ρ (dose, target score) is negative and significant for C1 rhythm complexity (-0.36), C3 transition validity (-0.33), C4 repetition adequacy (-0.52), C6 density/strain adequacy (-0.65), C7 density spike (-0.78), and C8 repetition adequacy (-0.42), with monotone mean trajectories (all $p < 10^{-15}$). C2’s target is the timing family (Section 5.3); C5’s designated band target holding steady is the anti-gaming property, reported next.

5.2 Reductio: borrowed metrics reward damage

Figure 1 shows the two reductio exhibits. *Perplexity rewards blandness:* official charts are more surprising (perplexity 7.98) than greedy machine output, and flattening human charts toward the LM’s modal patterns drives perplexity *down* by 31%. Read as “lower is better” [1], perplexity prefers the damaged chart—a corpus-LM instance of the likelihood/quality dissociation [10]. The band score built on the *same* LM statistic is not fooled ($0.892 \rightarrow 0.900$), because official charts sit near the band’s upper edge and flattening moves NLL toward the band center, not out of it; meanwhile the repetition-adequacy score catches the damage the attack actually does. This robustness survives a cross-LM version of the attack (drive on a held-half LM, measure on the full LM), upgrading it from an in-sample worst case to evidence that perplexity’s failure is not an artifact of attacking its own model.

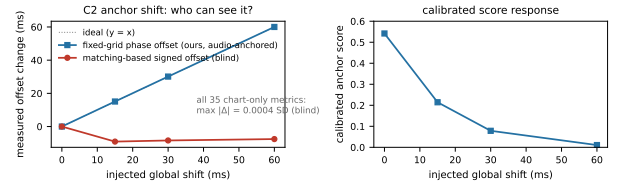


Figure 2. C2 anchor shift. **Left:** a constant $+15/30/60$ ms shift of every note against the audio is invisible to all 35 chart-only metrics (max response 0.0004 SD) and to the standard matching-based offset statistic, which re-matches shifted notes to neighboring anchors on a dense subdivision lattice (≈ 14 ms spacing at 180 BPM). The fixed-grid phase offset—support maximization on an eighth-note lattice anchored to a beat grid *estimated from audio*—recovers the injected shift to within 0.2 ms. **Right:** the calibrated anchor score collapses from 0.54 to 0.01 across doses. Results on the 129 charts whose estimated grid passes its own confidence gate; the estimated grid agrees with metadata grids on period exactly (31/31 songs) and on phase to ≈ 10 ms median.

Structureness rewards collapse: replacing charts with their own dominant 4-gram loop raises self-similarity structureness [9] monotonically—copy-paste is maximally self-similar—and set-level Self-BLEU rises in lockstep. On real systems the same failures appear in the wild: across eight generators spanning a known quality range, perplexity separates official from generated charts with AUC 0.32 (i.e., backwards), and several borrowed statistics are near-proxies for chart length (Spearman vs. note count up to 0.97, vs. ≤ 0.09 for the calibrated rankers).

5.3 Closing the anchor blind spot without references

A chart can be idiomatic on every chart-only axis and still be shifted off the music entirely—C2 makes this concrete (Figure 2). Two findings matter beyond the headline. First, the blindness extends to the obvious fix: a matching-based signed offset (match each note to its nearest meter anchor, average the residuals) is itself C2-blind, because a fine subdivision lattice re-matches every shifted note; detecting global misalignment requires estimating the phase that maximizes lattice support over a wrap-free window. Second, the anchors come from a beat grid estimated from audio by a beat-tracking model, not from any chart metadata, so the timing family is reference-free like the rest of the protocol; the estimation costs nothing measurable (period exact on all gated songs, phase agreement ≈ 10 ms, matching the estimator’s published precision). One honest caveat: a ≈ 15 ms grid-phase error on a single song can saturate that song’s unsupported-note rate, so per-song distributions (not means) should be reported and the phase-offset detector, not the unsupported rate, is the primary C2 signal.

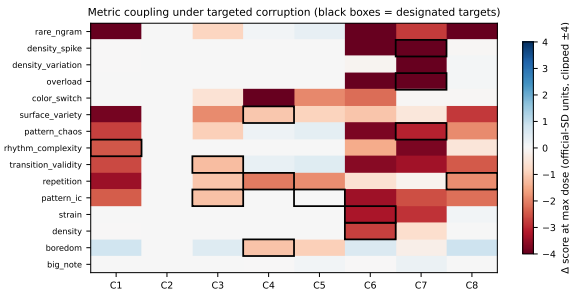


Figure 3. The full probe $\times$ score response matrix (official-SD units at max dose, clipped at ± 4 ; black boxes = designated targets). Targeted probes move their targets, but responses are *not* diagonal: timing jitter (C1) propagates into pattern scores because pattern tokens are IOI-quantized—in a rhythm game, timing damage *is* pattern damage—and the rare- n -gram diagnostic is hypersensitive to any token change. We publish the matrix as a coupling map rather than claiming orthogonality.

5.4 Coupling structure: what specificity we do not claim

We explicitly do not claim that each probe moves *only* its target. Figure 3 publishes the full response matrix: no probe passes a strict diagonal-dominance test, for two identifiable reasons. Shared tokenization couples axes—jittered timings change IOI bins and therefore every n -gram statistic downstream—which is partly domain truth (mistimed notes damage patterns as played) and partly a suite limitation. And one diagnostic-tier score (rare- n -gram rate) reacts to almost any token edit, which is why the taxonomy keeps it out of the ranker tier. Practitioners should read the suite as a coupled profile with a published response map, not as independent orthogonal axes.

5.5 Application: scoring real systems

Applied to eight generators from a controlled ablation family of conditional chart models (grid-conditioned and gridless decoding modes; two configurations whose training-budget allocation is known to starve the gridless task, F-measure 0.58 and 0.36 vs. 0.66–0.67 for healthy configurations), the calibrated profile reproduces the known ordering with graded response (chart-quality profile 0.83–0.86 healthy, 0.79 semi-collapsed, 0.68 collapsed; official 0.916), while the audio-anchored family exposes a failure no chart-only score sees: the collapsed system carries a 38.5ms systematic offset against the music (healthy systems ≈ 10 ms). Because calibration bands are fixed corpus artifacts, scores are comparable across studies: re-scoring a two-generations-older system family under the same bands shows transition validity rising from 0.37 to 0.75–0.82—a cross-generation progress statement that single-reference F-measure cannot make. Two dissociations illustrate what F-measure misses: a dedicated gridless model scores higher onset F1 than a multi-task model (0.673 vs. 0.659) yet lower transition validity (0.753 vs. 0.785–0.791), and greedy decoding pushes band-adequacy

scores *above* official values on some axes—band scores reward in-distribution blandness and must be read jointly with the variety/repetition axes (which is what the profile is for).

6. LIMITATIONS AND FUTURE WORK

No human validation. The probes certify damage detection, not experience prediction; validating the surviving metrics against player judgments remains future work. **Population-level direction.** 4–14% of individual charts move toward their band under corruption; per-chart verdicts need the profile, not one score. **Coupling.** The suite is a coupled profile (Section 5.4); attribution between timing and pattern damage is smeared by shared tokenization. **Scope.** One game domain for the calibration corpus (the cross-game study covers only the game-agnostic families), greedy decoding for the real-system study, and course-level (not level-conditioned) calibration bands. **Ceiling.** Among healthy systems the profile differences sit within evaluation noise; distinguishing good from slightly-better remains a human question.

7. CONCLUSION

Chart generation needs evaluation it can trust more than it needs another metric. Corruption probes make metric validation systematic and reproducible: the borrowed metrics did not merely underperform—they rewarded blandness, rewarded collapse, and missed a fatal global misalignment entirely. Corpus-calibrated band scores over the same raw quantities, extended with an audio-anchored timing family, pass the same probes with monotone dose response and sub-ms shift recovery, with their coupling structure published rather than hidden. The harness, calibration artifacts, per-chart records, and evaluation package are released so that future chart-quality metrics can be validated under the same protocol. [PENDING: P8: artifact URL.]

8. ETHICS STATEMENT

The corpus consists of community-created charts used under research terms; no personal data is involved. Better automated evaluation chiefly benefits players and small developers; we see no direct misuse vector beyond automated content farming, which platform moderation already addresses.

9. AI USAGE STATEMENT

AI assistants were used for experiment orchestration, drafting, and editing under author direction; all numbers were produced by the released code, adversarially re-derived, and verified by the authors.

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